Summary in Graph

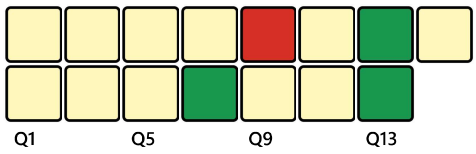
Exam Summary (GO Classes CS/DA Test Series 2026 | Databases | Topic Wise Test 3)

Qs. Attempted:	**6** (1 + 5)	**Correct Marks:**	**6** (0 + 6)
Correct Attempts:	**3** (0 + 3)	**Penalty Marks:**	**0.67** (0 + 0.67)
Incorrect Attempts:	**3** (1 + 2)	**Resultant Marks:**	**5.33** (0 + 5.33)

Total Questions:	**15** (5 + 10)
Total Marks:	**25** (5 + 20)
Exam Duration:	**45 Minutes**
Time Taken:	**45 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (153) out of 1532 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Suppose that two relations $A(X, Y)$ and $B(X, Y)$ have exactly the same schema. Also, there are no null values in the relations.
Which of the following equalities hold in relational algebra? (Where " $\times$ " is Cartesian-Product Operation)

A. $A \cap B = A - (A - B)$
B. $A \cap B = B - (B - A)$

C. $A \cap B = A$ NATURAL-JOIN B
D. $A \cap B = A \times B$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Databases

Consider the following relation schemas: r(A, B), t(B, C) and their instances :

$$r(A, B) = \{\langle a, b\rangle, \langle a, c\rangle, \langle c, d\rangle, \langle b, e\rangle\}$$
$$t(B, C) = \{\langle b, e\rangle, \langle d, h\rangle, \langle b, f\rangle, \langle a, d\rangle, \langle a, e\rangle\}$$

How many rows are present in the result of the following relational algebra expression using left-outer-natural-join :

$$\pi_{A, C}\ (r \bowtie_{\text{LEFT}} t)$$

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Databases

Consider the following relations and query.

Use Venn Diagram

$$r(A, B, C)$$
$$s(A, B, C)$$
$$\sigma_{A>C}([(r \cup s) - (r - s)] - (s - r))$$

Which of the following is equivalent to the above query?

A. $\sigma_{A>C}(r \bowtie s)$
B. $\sigma_{A>C}(r \cup s)$
C. $\sigma_{A>C}(r) \cap \sigma_{A>C}(s)$
D. $\sigma_{A>C}(r \cap s)$

Your Answer: A;B;D | Correct Answer: A;C;D | Incorrect | Time taken: 03min 20sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the relations r, s and query :

$$t1 \leftarrow \sigma_\theta((r \times s) \div r)$$
$$t2 \leftarrow \sigma_\theta(s)$$
$$t3 \leftarrow \sigma_\theta(r)$$

Relation r is not empty.

Which of the following is correct?

A. $t1 = t3$
B. $t2 = t3$
C. $t1 = t2$
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Consider the following schema:

Sailors(*sid:* integer, *sname:* string, *rating:* integer, *age:* real)

Boats(*bid:* integer, *bname:* string, *color:* string)

Reserves(*sid:* integer, *bid:* integer, *day:* date)

The primary key fields are underlined, and the domain of each field is listed after the field name. Assume there are no NULL values in any column of any relation.

Consider the following query :

```
SELECT S.sname
FROM Sailors S, Reserves R, Boats B
WHERE S.sid = R.sid AND R.bid = B.bid
    AND (B.color = 'blue' AND B.color = 'white')
```

What is the output of the above query?

A. The names of sailors who have reserved a blue or a white boat.
B. The names of sailors who have reserved a blue AND a white boat.
C. The names of sailors who have reserved neither a blue nor a white boat.
D. Empty set.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following Student database and the relations in it :

Student (sid, name, marks)
Course (cid, min, max)

Student relation has the following set of tuples :

$$\{(101, a, 50), (102, b, 52), (103, c, 60), (104, d, 70)\}$$

Course relation has the following set of tuples :

$$\{(c1, 20, 90), (c2, 15, 95), (c3, 10, 100)\}$$



Consider the following query :

```
SELECT S.sid
FROM Student S
WHERE EXISTS ( SELECT COUNT(*)
               FROM Course C
5.             WHERE S.marks > C.min AND S.marks > C.max)
```

How many tuples will be in the output of the above SQL Query?

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 03min 38sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Let $\mathrm{R} = (\mathrm{A}, \mathrm{B}, \mathrm{C}), \mathrm{S} = (\mathrm{C}, \mathrm{D}, \mathrm{E})$ and let q and r be a relation on schema R and s be a relation on schema S.

Consider the following queries :

1. $\{< a > | \exists b(< a, b > \in r \wedge b = 10)\}$
2.
```
SELECT DISTINCT A FROM r WHERE B=10
```
3.
```
SELECT DISTINCT * FROM r EXCEPT SELECT DISTINCT * FROM s
```
4.
```
SELECT DISTINCT * FROM q WHERE (A, B, C) NOT IN (SELECT DISTINCT * FROM r)
```

Which of the following is/are true? (Two queries are said to be equivalent when they produce the same output, i.e., the same set of records)

A. Query 1 and 2 are equivalent.
B. Query 1 and 3 are equivalent
C. Query 2 and 4 are equivalent
D. Query 3 and 4 are equivalent

Your Answer: A;B;C | Correct Answer: A | Incorrect | Time taken: 04min 13sec | Discuss

Q #8 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Consider the following database relations :

- Enrolled(Student, Course) : It contains the courses that each student has enrolled into, and
- ToDo(Course) : It contains the courses that are required to graduate.

The instance for Enrolled relation is shown below:

Student	Course
Robert	Databases
Robert	Programming Languages
Susie	Databases
Susie	Operating Systems
Julie	Programming Languages
Julie	Machine Learning
Emilie	Operating Systems

The instance for ToDo relation is shown below:

Course
Databases
Programming Languages

We run the following SQL Query of this database :

```
SELECT DISTINCT Student
FROM Enrolled
WHERE Course IN (SELECT Course FROM ToDo);
```

How many tuples will be there in the output of this SQL Query?

Your Answer: 3 | Correct Answer: 3 | Correct | Time taken: 02min 58sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following schema:

Sailors(*sid:* integer, *sname:* string, *rating:* integer, *age:* real)

Boats(*bid:* integer, *bname:* string, *color:* string)

Reserves(*sid:* integer, *bid:* integer, *day:* date)

The primary key fields are underlined, and the domain of each field is listed after the field name. Assume there are no NULL values in any column of any relation and every relation has at least one tuple. "Sid" of Reserves is foreign key to Sailor.sid.
Assume that there exists a sailor who has reserved a boat with bid 104 in the reserves table.

Consider the following query :

```
SELECT S.sid
FROM Sailors S, Reserves R
WHERE S.rating =10 OR R.bid =104
```

What is the output of the above query?

A. All sids of sailors who have a rating of 10 AND have reserved boat 104.
B. All sids of sailors who have a rating of 10 OR have reserved boat 104.
C. All sids of sailor table
D. Empty set.

Your Answer: B | Correct Answer: C | Incorrect | Time taken: 07min 05sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Assume we have two relations $\mathrm{R}(\mathrm{A}, \mathrm{B})$ and $\mathrm{S}(\mathrm{B}, \mathrm{C})$. All three attributes $(\mathrm{A}, \mathrm{B}, \mathrm{C})$ are integer attributes. Assume that Relation R contains the following tuples: $(1,2),(2,3)$, and $(3,4)$. Assume that relation S contains the following tuples $(2,2),(2,3)$, and $(5,1)$. Note that " $\times$ " is Natural join, in the following expressions.

Which of the following relational algebra expressions are equivalent, with respect to the above information?

A. $\pi_{\mathrm{A}}\left(\sigma_{\mathrm{B}=1}(\mathrm{R})\right)$ and $\sigma_{\mathrm{B}=1}\left(\pi_{\mathrm{A}}(\mathrm{R})\right)$
B. $\pi_{\mathrm{A}}\left(\sigma_{\mathrm{A}=1}(\mathrm{R})\right)$ and $\sigma_{\mathrm{A}=1}\left(\pi_{\mathrm{A}}(\mathrm{R})\right)$
C. $\pi_{\mathrm{A}, \mathrm{B}}(\mathrm{R} \times \mathrm{S})$ and $\left(\pi_{\mathrm{A}, \mathrm{B}}(\mathrm{R})\right) \times \mathrm{S}$
D. $\sigma_{\mathrm{B}=1}(\mathrm{R} \times \mathrm{S})$ and $\left(\sigma_{\mathrm{B}=1}(\mathrm{R})\right) \times \mathrm{S}$

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Given relations r and s. Let Function C denote the number of results in the output. Which of the following statements "can be" correct? (Hint: Discard the statements which are NEVER possible)

I $C(r \bowtie s) > C(r \rhd\!\bowtie s)$
II $C(r \rhd\!\bowtie s) > C(r \rhd\!\!\rhd s)$
III $C(r \rhd\!\!\rhd s) > C(r \bowtie s)$
IV $C(r \rhd\!\bowtie s) > C(r \bowtie s)$

A. I and II
B. II and III
C. Only III
D. III and IV

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Databases

For a simple undirected finite graph G, graph degrees (degrees of vertices of graph) are stored in a database table $\mathrm{Deg(G)}$. This table has two attributes V, D where V represents the vertices and D represents corresponding degree of those vertices i.e. If v_i is some vertex of degree d_i in graph G then there will be a row v_i, d_i in $\mathrm{Deg(G)}$. V is the primary key in this table. For a 100 vertices simple graph G, the summation of minimum and maximum possible value of $|\pi_{\mathrm{D}}(\mathrm{Deg(G)})|$ will be _____

Your Answer: | Correct Answer: 100 | Not Attempted | Time taken: 03min 52sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Consider the following three relational schemas in a database :

```
CREATE TABLE R(A INT PRIMARY KEY, B INT);
```

```
CREATE TABLE S(C INT PRIMARY KEY, A INT REFERENCES R(A) ON DELETE CASCADE);
```

```
CREATE TABLE T(D INT PRIMARY KEY, C INT REFERENCES S(C) ON DELETE SET NULL);
```

Consider the following instances of relations R, S, T :

R

A	B
a	a
b	b

S

C	A
b	a
c	a

T

D	C
a	b
c	c

Which of the following is correct?

A. After executing the command "DELETE FROM T;", R will remain unchanged.
B. After executing the command "DELETE FROM S;", R will remain unchanged.
C. After executing the command "DELETE FROM S;", T will contain 2 tuples.
D. After executing the command "DELETE FROM R;", T will contain 2 tuples.

Your Answer: A;B;C;D | Correct Answer: A;B;C;D | Correct | Time taken: 03min 48sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Let $\mathrm{R}(a, b, c, d)$ be a RDBMS relation having the following set of functional dependencies(FD set) :
FD Set $\mathrm{F} = \{b \rightarrow c; c \rightarrow d; ab \rightarrow d; bc \rightarrow cd\}$ We apply some queries on a relational instance r of R. Which of the following queries are guaranteed to return the same number of tuples as in relation instance r?

A. $\prod_{a}(r)$
B. $\prod_{a,c}(r)$
C. $\prod_{a,b}(r)$
D. $\prod_{a,b,d}(r)$

Your Answer: C;D | Correct Answer: C;D | Correct | Time taken: 16min 14sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

For a RDBMS relation $\mathrm{T}(\mathrm{P}, \mathrm{Q}, \mathrm{R}, \mathrm{S}, \mathrm{U})$, the following set F of functional dependencies holds:

$$\mathrm{F} = \{\mathrm{P} \rightarrow \mathrm{QR}; \mathrm{RS} \rightarrow \mathrm{U}; \mathrm{Q} \rightarrow \mathrm{S}; \mathrm{U} \rightarrow \mathrm{P}\}$$

- Ram decomposes T into two relations $\mathrm{X}(\mathrm{P}, \mathrm{Q}, \mathrm{R}), \mathrm{Y}(\mathrm{R}, \mathrm{S}, \mathrm{U})$.
- Shyam decomposes T into two relations $\mathrm{A}(\mathrm{P}, \mathrm{Q}, \mathrm{R}), \mathrm{B}(\mathrm{P}, \mathrm{S}, \mathrm{U})$.

Which of the following is/are necessarily true on any(every) instance t of relation T?

A. $\mathrm{X} \bowtie \mathrm{Y} = t$
B. $\mathrm{X} \bowtie \mathrm{Y} \neq t$
C. $\mathrm{A} \bowtie \mathrm{B} = t$
D. $\mathrm{A} \bowtie \mathrm{B} \neq t$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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